

dental_clinic_db — Logical Schema & Data Dictionary

Logical Schema

clinic_clients (client_id [PK], first_name, last_name, dob, phone_number, email_address, registered_date)

practitioners (doctor_id [PK], doctor_fname, doctor_lname, specialty, license_no, hire_date)

treatments (treatment_code [PK], treatment_name, category, base_cost, price_updated_date)

patient_visits (visit_id [PK], client_id [FK → clinic_clients.client_id], doctor_id [FK → practitioners.doctor_id], scheduled_date, scheduled_time, booking_date, visit_status)

visit_procedures (entry_id [PK], visit_id [FK → patient_visits.visit_id], treatment_code [FK → treatments.treatment_code], target_tooth, status, billed_cost)

Data Dictionary

1. clinic_clients

Column	Type	Constraints	Description
client_id	INT	PK, AUTO_INCREMENT	Unique ID for each patient
first_name	VARCHAR(50)	NOT NULL	Client's given name
last_name	VARCHAR(50)	NOT NULL	Client's family name
dob	DATE	NOT NULL	Patient's date of birth
phone_number	VARCHAR(20)	NOT NULL	Primary contact number
email_address	VARCHAR(100)	NULLABLE, UNIQUE	Patient's email address
registered_date	DATE	NOT NULL, DEFAULT CURRENT_DATE	Date client first enrolled

2. practitioners

Column	Type	Constraints	Description
doctor_id	INT	PK, AUTO_INCREMENT	Unique doctor identifier
doctor_fname	VARCHAR(50)	NOT NULL	Doctor's first name
doctor_lname	VARCHAR(50)	NOT NULL	Doctor's last name
specialty	VARCHAR(80)	DEFAULT 'General Dentistry'	Area of expertise
license_no	VARCHAR(40)	NOT NULL, UNIQUE	Official license code
hire_date	DATE	NOT NULL	Date doctor joined the practice

3. treatments

Column	Type	Constraints	Description
treatment_code	INT	PK, AUTO_INCREMENT	Unique treatment code
treatment_name	VARCHAR(100)	NOT NULL	Procedure title
category	VARCHAR(50)	NOT NULL	Preventive, Cosmetic, Surgical
base_cost	DECIMAL(10,2)	NOT NULL, CHECK (base_cost >= 0)	Default catalog price

Column	Type	Constraints	Description
price_updated_date	DATE	NOT NULL	Date fee was last updated

4. patient_visits

Column	Type	Constraints	Description
visit_id	INT	PK, AUTO_INCREMENT	Unique appointment identifier
client_id	INT	NOT NULL, FK → clinic_clients(client_id), ON DELETE RESTRICT	Attending client
doctor_id	INT	NOT NULL, FK → practitioners(doctor_id), ON DELETE RESTRICT	Assigned practitioner
scheduled_date	DATE	NOT NULL	Reserved date for the visit
scheduled_time	TIME	NOT NULL	Scheduled time slot
booking_date	DATE	NOT NULL, DEFAULT CURRENT_DATE	Date reservation was created
visit_status	VARCHAR(20)	NOT NULL, DEFAULT 'Scheduled', CHECK IN ('Scheduled','Completed','Cancelled','No-Show')	Visit state

Additional constraint: UNIQUE (doctor_id, scheduled_date, scheduled_time) — prevents double-booking a doctor.

5. visit_procedures

Column	Type	Constraints	Description
entry_id	INT	PK, AUTO_INCREMENT	Unique procedure entry ID
visit_id	INT	NOT NULL, FK → patient_visits(visit_id), ON DELETE RESTRICT	Parent visit record
treatment_code	INT	NOT NULL, FK → treatments(treatment_code), ON DELETE RESTRICT	Treatment planned/executed
target_tooth	INT	NULLABLE, CHECK (target_tooth BETWEEN 1 AND 32)	Tooth number (Universal Numbering)
status	VARCHAR(20)	NOT NULL, DEFAULT 'Pending', CHECK IN ('Pending','Completed','Cancelled')	Procedure status
billed_cost	DECIMAL(10,2)	NULLABLE, CHECK (billed_cost >= 0)	Final charged price (NULL while Pending)